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EDUCATION

Rochester Institute of Technology

Master of Science in Computer Science – Specialization in Distributed Systems

Aug 2024 – Dec 2025

GPA: 4/4

University of Pune

Bachelor of Engineering in Computer Engineering

Aug 2019 – Aug 2022

GPA: 3.85/4

University of Mumbai

Diploma of Engineering in Computer Engineering

Aug 2016 – Jun 2019

GPA: 3.64/4

SKILLS

Languages: Python, Java, C++, JavaScript (ES6+), HTML/CSS, Bootstrap, SQL, AngularJS

Frameworks & Libraries: Django, FastAPI, REST APIs, TensorFlow, Scikit-learn, NumPy, Pandas, OpenCV

Databases & Messaging: MongoDB, MySQL, PostgreSQL, Elasticsearch, Redis, Kafka, RabbitMQ, ZooKeeper

Cloud & DevOps: AWS, Docker, Kubernetes, Kibana, Git, BitBucket, CI/CD (GitHub Actions, Jenkins)

Tools & Methodologies: Visual Studio Code, Android Studio, Postman, JSON, Linux, gRPC, Microservices, OOP

Networking & Protocols: TCP/IP, IPv4, IPv6, HTTP, DNS, SSL/TLS, WebSockets, Load Balancing, Wireshark

Project Management: JIRA, Agile Methodologies

EXPERIENCE

Software Engineer – Quick Heal Technologies Ltd

Jan 2023 – Jul 2024

CI/CD Pipeline Enhancement | *Python, Java, Redis, Celery, Docker, MongoDB, Kibana, Elasticsearch*

- Streamlined a critical CI/CD pipeline with Python, achieving 60% faster deployments and higher reliability.
- Implemented Redis-backed Celery task queues, cutting API latency by 40%, added MongoDB/ElasticSearch monitoring to reduce root-cause analysis time by 50%

Ambrose Portal for Centralized Project Processing | *Python, FastAPI, PostgreSQL, MongoDB, JavaScript*

- Led the development of a centralized portal using FastAPI, PostgreSQL, React-like front-end for managing and processing projects efficiently, with a focus on usability and scalability.
- Designed and implemented a multithreaded, asynchronous API backend, enabling real-time processing of 100+ concurrent requests daily with 0 downtime, improving system stability under peak load by 60%.
- Reduced average processing time from 10 hours to 2 hours per week by implementing a scalable microservices architecture, automating workflows, and monitoring system performance through Grafana dashboards, improving team efficiency by 80%.

Expiro File Infector Virus Recovery Tool | *Java, Python, MongoDB, Flask*

- Engineered a comprehensive solution for detecting and recovering from file infections, leveraging feedback from 5 key stakeholders during sprint cycles.
- Designed and maintained REST APIs(Flask), microservices (Python) and recovery heuristics (hash-based integrity + repair rules) over MongoDB; processed 1,00,000 files/day with 95% successful recovery.

Signature Matching Language (SML) Portal | *Django, Python, MySQL, HTML/CSS, Bootstrap, JavaScript*

- Shipped an internal SML authoring and lifecycle portal for 50+ threat researchers (create → test → deploy), replacing manual scripts and email handoffs, cutting signature deployment errors by 80%.
- Built a versioned rules store and review workflow in Django/MySQL with RBAC, audit logs, and one-click promotion to the AV engine; cut signature release lead time from 12 hours to 3 hours (-75%).

Graduate Teaching Assistant – Rochester Institute of Technology

Aug 2025 – Present

- Provided technical mentorship to 85+ students in Python programming and CS Theory via weekly labs, office hours, and rubric-based grading; delivered fast, actionable feedback that improved clarity and outcomes.

PROJECTS

Federated Learning + Blockchain for ECG Anomaly Detection | *Python, Blockchain, Solidity, Ganache* **2024**

- Engineered a decentralized Federated Learning pipeline using DTW-based clustering and weighted FedAvg, integrated with Solidity smart contracts for secure, auditable, and privacy-preserving ECG model updates.

Realtime Road Lane Detection | *Python, ML, Linux, OpenCV, TensorFlow, MQTT* | [Research Paper](#) **2022**

- Architected a real-time road lane detection system using OpenCV, processing 60 FPS video with grayscale filtering, ROI masking, and Canny Edge Detection to identify lane contours under varying road and lighting conditions.
- Deployed the edge pipeline on Linux and streamed deviation alerts via MQTT for instant driver feedback.